AgentMorph — Project Execution Plan

Solo researcher on Colab Pro+ with Claude and Claude Code · Target: NeurIPS 2026 D&B

Stage & Timeline	What we do	How we do it (with Claude / Claude Code)	Outcome
Stage 1 — Foundation	Set up the agent test-bed. Pick 5 small open models (Llama-3.2-3B, Qwen2.5-7B, Gemma-2-9B, Phi-4, Llama-3.1-8B). Integrate with smolagents + LangGraph. Load AgentDojo and build a synthetic 30-tool e-commerce suite.	Claude Code writes the agent wrappers, tool definitions, and orchestration scripts. Claude (chat) helps plan the tool schema for the synthetic suite. All 5 models run 4-bit quantized on Colab T4	Working pipeline: any of 5 models × 2 frameworks × 2 environments, reproducible end-to-end. ~200 baseline trajectories logged.
Stage 2 — Rule Library + Mutation Engine	Define the ~25 metamorphic rules (tool-order invariance, schema paraphrase, read-only idempotency, synonym robustness, refusal consistency, etc.). Build a mutation engine that applies each rule to any agent scenario.	Claude (chat) helps draft the rule catalog grounded in the MAST failure taxonomy. Claude Code implements each mutator as a Python module with unit tests. LLM-guided mutation uses Gemini 2.5 Flash (free tier) for paraphrasing.	agentmorph-core Python library: 25 mutators + equivalence checkers, all unit-tested. Ready to plug into any agent.
Stage 3 — Large-Scale Experiments + Transfer	Run all 5 open models × 25 rules × ~100 scenarios ≈ 12,500 mutation pairs. Log every violation. Then run a 300-bug transfer study on Groq Llama-3.3-70B, Gemini 2.5 Flash, Mistral Small, and Claude Haiku starter credits.	Claude Code writes the experiment runner with resumable checkpoints (Colab sessions die) and aggressive KV-cache reuse. Claude (chat) debugs failures, designs the violation-logging schema, and helps write analysis notebooks.	~1,500 labeled agent bugs, per-model reliability leaderboard, transfer-rate tables. All raw logs ready for release.
Stage 4 — Human Validation, Packaging, Paper	Human-validate 100 sampled bugs (recruit 2–3 labmates, ~weekend effort). Package agentmorph as a pip-installable library. Release the bug dataset on HuggingFace. Write and submit the paper.	Claude drafts README, docstrings, dataset card, and the full NeurIPS D&B paper section-by-section. Claude Code automates pip packaging, CI tests, and figure generation. Human validation handled manually in a shared spreadsheet.	Public library + HF dataset + 9-page NeurIPS 2026 D&B submission (deadline ~4 may 2026). Fallback: ICLR 2027 workshops, IEEE TSE.